

RAHUL RAY

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EDUCATION

B.Tech, Computer Science and Engineering, Assam University
CGPA: 7.43 / 10

2019 – 2023

TECHNICAL SKILLS

Languages/Frameworks: Python (Advanced), SQL, FastAPI, Pydantic, Laravel, Java, C++
Cloud & DevOps: Docker, AWS (EC2, S3, IAM), Git, CI/CD, Linux (Bash), Docker Compose
Data & AI: MySQL, PostgreSQL, ChromaDB (Vector), Redis, RAG Systems, LangChain
AI Tooling/DX: Cursor (AI IDE), Windsurf, CodeRabbit, Postman, VS Code

PROFESSIONAL EXPERIENCE

Software Engineer (Young Professional-II, IT)

Dec 2024 – Present

ICAR - Indian Agricultural Statistics Research Institute (IASRI), New Delhi

- Architected backend services for ASHOKA HPC Hub, designing RESTful APIs managing access to a 1.5 PB genomic data lake for the National Agricultural Bioinformatics Grid (NABG).
- Led migration of data processing modules from Laravel to asynchronous FastAPI microservices, significantly improving throughput for I/O-heavy tasks.
- Optimized MySQL schemas with composite indexing strategies, eliminating full-table scans and reducing query latency for multi-terabyte datasets.
- Implemented JWT authentication and RBAC for a secure API layer, enabling mobile-to-cloud data synchronization for field researchers.

Independent Backend Engineering Researcher

Aug 2023 – Nov 2024

Remote / Self-Directed

- **Enterprise RAG System:** Built a production-ready Q&A API using FastAPI and Docker with asynchronous document ingestion into ChromaDB; deployed on AWS (EC2, S3) with Nginx reverse proxy.
- **Scalargrad Engine:** Implemented a custom Python autograd engine from first principles using Directed Acyclic Graphs (DAGs) for gradient backpropagation; validated correctness against PyTorch.

Android Development Intern

Jun 2022 – Jul 2022

IIT BHU, India

- Built a real-time news aggregation app in Java using Retrofit; improved performance and UX by integrating Picasso for asynchronous image loading and caching.

TECHNICAL PROJECTS

Raft Consensus Algorithm | *Python, FastAPI, Distributed Systems*

May 2026

- Implemented Raft consensus from scratch in Python — leader election with randomized timeouts, log replication with majority quorum commits, and automatic failover on leader crash.
- Built TCP/HTTP transport layer using FastAPI for inter-node communication and a live browser visualization showing node states, term numbers, and state machine in real time.

Gossip Protocol: Peer-to-Peer Failure Detector | *Python, UDP Sockets, Threading, Distributed Systems*

Apr 2026

- Built gossip-based distributed failure detector with epidemic propagation over UDP, decentralised membership tracking, eventual consistency via heartbeat merge, and sub-15s failure detection across N nodes converging in $O(\log N)$ rounds.